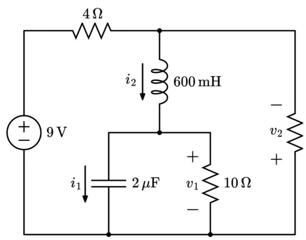
Practice Quiz Seven: RLC Circuit Analysis under DC Conditions (Easy)

For the RLC circuit at steady state shown below, find the following:

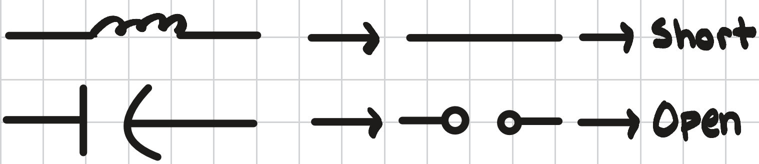
a) v_1
b) v_2
c) i_1
d) i_2

Finally, find the energy stored in both the inductor and the capacitor.

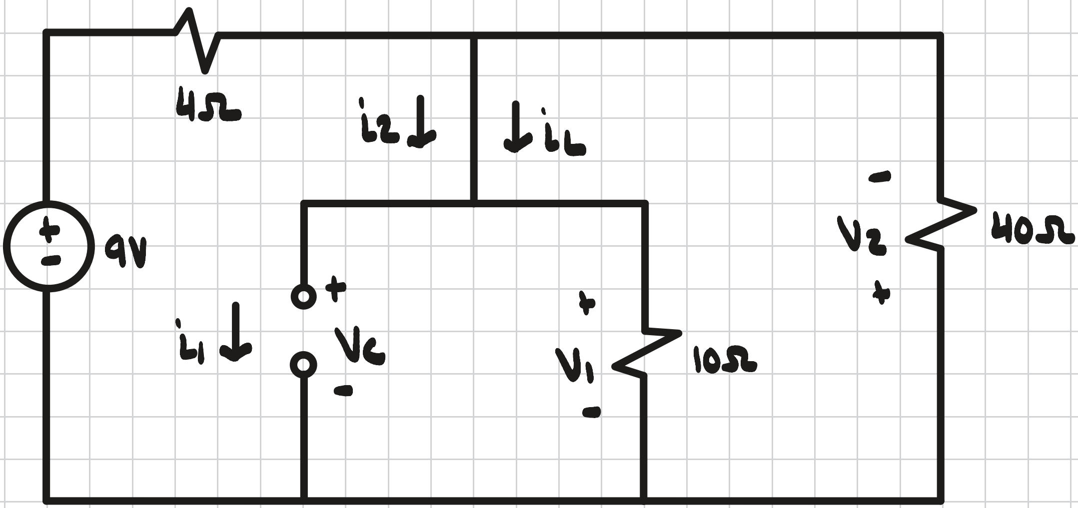
Suggested time limit: 10 minutes



Steady State Conditions



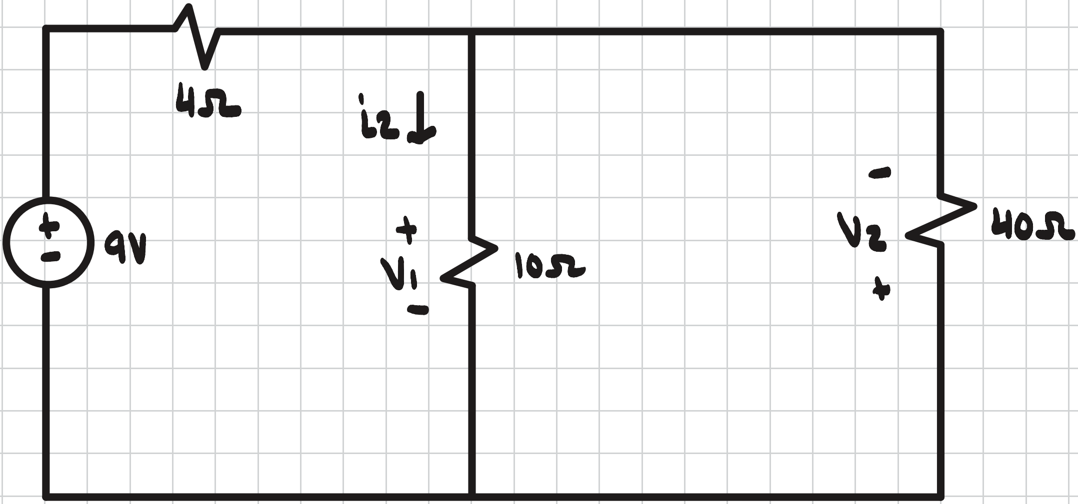
Step #1 Redraw Circuit



Step #2 Find i_1

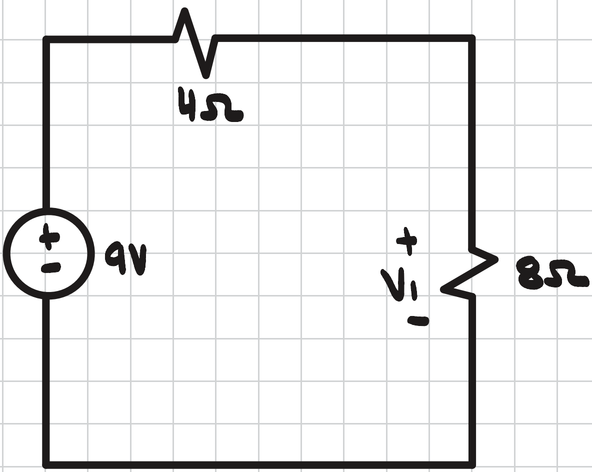
$i_1 = 0A$

Step #3 Redraw Circuit



Step #4 Find V_1

$$\left(\frac{1}{40} + \frac{1}{10}\right)^{-1} = 8\Omega$$

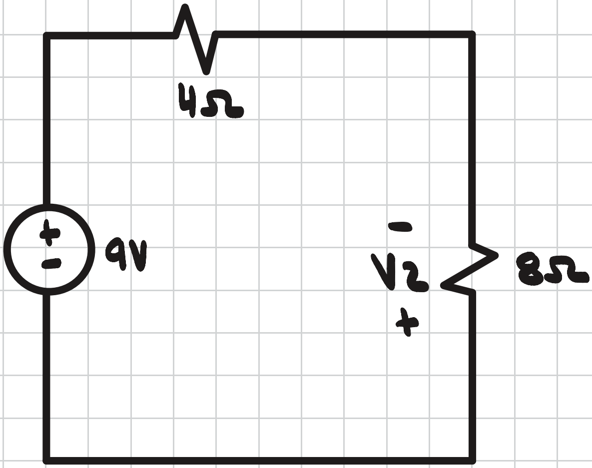


$$V_1 = 9\left(\frac{8}{8+4}\right)$$

$V_1 = 6V = V_c$

Step # 5 Find V_2

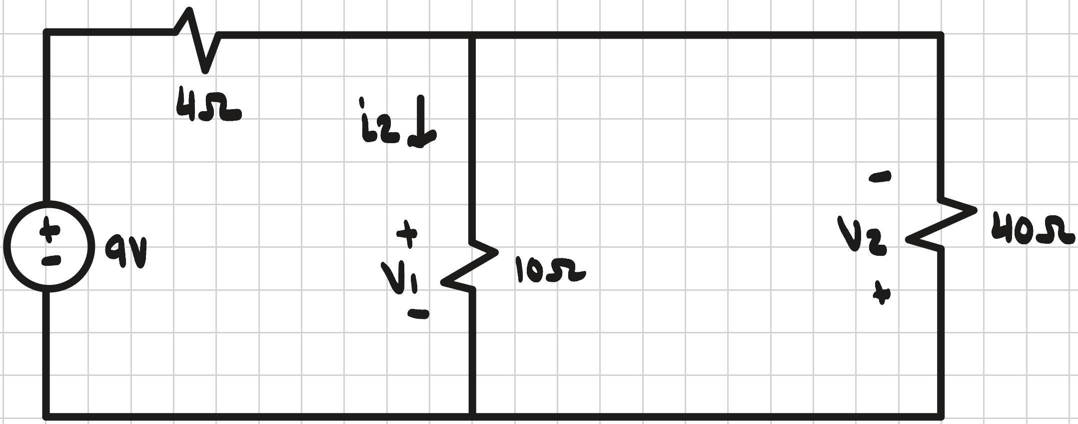
$$\left(\frac{1}{40}+\frac{1}{10}\right)^{-1}=8\,\Omega$$



$$V_2=-9\left(\frac{8}{8+4}\right)$$

$$\boxed{V_2=-6\,V}$$

Step #6 Find i_2



$$i_2=\frac{V_1}{10}=\frac{6}{10}$$

$$\boxed{i_2=600\,mA=i_L}$$

Step # 7 Find Energy

$$W_{capacitor} = \frac{1}{2} CV^2 = \frac{1}{2}(2 \cdot 10^{-6})(6)^2$$

$$W_{capacitor} = 36 \mu J$$

$$W_{Inductor} = \frac{1}{2} L i_L^2 = \frac{1}{2}(600 \cdot 10^{-3})(600 \cdot 10^{-3})^2$$

$$W_{Inductor} = 0.108 J$$